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PAPER-1 (COMPULSORY)

TEACHING AND RESEARCH APTITUDE

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GENERAL PAPER

[10 JAN 2025 (M)]

Comprehension: The following table shows the number of students enrolled in four different Colleges (A-D) during the four years from 2020 to 2023. Based on the data in the table, answer the questions that follow.

Year-wise Enrolment of Students in Colleges				
Year	Number of Students Enrolled in College			
	A	B	C	D
2020	1200	900	1000	700
2021	1100	800	1300	900
2022	1400	1000	1200	1100
2023	600	700	800	1000

- Number of students enrolled in College A in year 2020 and 2021 together is _____% more than the number of students enrolled in College C in the year 2022 and 2023 together.
 - 12.5
 - 8.5
 - 10
 - 15
- If the ratio between number of girls and boys in College B in 2021 is 3 : 5 and the ratio between the number of girls and boys in College C in 2022 is 7 : 5, then what is the ratio between the number of girls in College B in 2021 to the number of boys of College C in 2022?
 - 2 : 5
 - 3 : 5
 - 1 : 3
 - 2 : 3
- Taking all the Colleges together, the total number of students enrolled in the year 2020 is _____ more than the total number of students enrolled in the year 2023.
 - 500
 - 900
 - 300
 - 700
- What is the average number of students enrolled in College B over all the four years?
 - 850
 - 750
 - 840
 - 720
- For College B in the year 2021, if 20% of students were enrolled in B.Sc, 30% of students were enrolled in B.Com. and the remaining students were enrolled in BCA, then what is the difference between the number of students enrolled in BCA and B.Sc?
 - 260
 - 220
 - 230
 - 240

- Identify the advantages of video conferencing.

- Increased productivity
- Going green
- Lack of personal contact
- Clarity of audio and video always

Choose the **correct** answer from the options given below:

- A, B and D Only
 - A and C only
 - A, B and C Only
 - A and B Only
- A sum of Rs 3800 is divided among three persons A, B, and C in the ratio $1/2 : 1/3 : 3/4$ what is the share of the person C?
 - Rs. 3360
 - Rs. 2240
 - Rs. 2150
 - Rs. 1800
 - Which of the following soil type supports earth worm activity?
 - Loam soil
 - Gleyed soil
 - Rendzina
 - Podzol

Choose the **correct** answer from the options given below:

- A, B and C Only
 - B, C and D only
 - A, C and D Only
 - A, B and D Only
- Arrange the following OSI layers in correct sequence:
 - Data link layer
 - Physical layer
 - Network layer
 - Presentation layer
 - Application layer

Choose the **correct** answer from the options given below:

- E, D, B, A, C
- E, D, A, C, B
- B, A, C, E, D
- B, A, C, D, E

- Match the **List-I** with **List-II**.

List-I (Proponents)		List-II (Theory)	
A.	George Gerbner	I.	Cognitive Dissonance Theory
B.	L.A. Festinger	II.	Cultivation Theory
C.	McCombs and Donald L. Shaw	III.	Uses and Gratification Theory
D.	Katz and Blumer	IV.	Agenda Setting Theory

Choose the **correct** answer from the options given below:

- (1) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- (2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
- (3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
- (4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

11. Match the **List-I** with **List-II**.

List-I (Institutions)		List-II (Propounders)	
A.	Milind College	I.	B.R. Ambedkar
B.	Rajghat Besant School	II.	Sri Aurobindo
C.	Banasthali Vidyapeeth	III.	Hiralal Shastri
D.	Auroville	IV.	Jiddu Krishnamurti

Choose the **correct** answer from the options given below:

- (1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- (2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
- (3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
- (4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

12. Which of the following statements are correct in the context of operating system?

- (A) Utility program is the first program loaded into the computer during booting
- (B) Multiprogramming was introduced for interleaved execution of multiple jobs by the same CPU
- (C) DOS is an example of single user operating system.
- (D) Mainframe computer uses multiuser operating system.

Choose the **correct** answer from the options given below:

- (1) A, B and C Only
- (2) A, B and D Only
- (3) B, C and D Only
- (4) A, C and D Only

13. Which of the following statements could be considered as a valid scientific hypothesis?

- (1) Adults should drink more water.
- (2) Eating 20 almonds a day decreases the risk of contracting heart disease
- (3) Cows should be given green fodder
- (4) Children should eat cereals

14. The infamous minutes of 1834 can be associated with

- (1) Charles Wood
- (2) William Bentuick
- (3) Thomas Babington Macaulay
- (4) Charles Grant

15. Individual and group accountability is one of the elements of _____.

- (1) Inductive learning
- (2) Cooperative learning
- (3) Deductive learning
- (4) Active learning

16. Which of the following statements is true for Episodic Buffer?

- (1) Integrates information from phonological loop only.
- (2) Integrates information from visuospatial sketch pad only.
- (3) Integrates information from the phonological loop and visuospatial sketchpad.
- (4) Integrates information from the phonological loop, visuospatial sketchpad and long-term Memory.

17. Which school among the following upholds perception as an unerring knowledge of the unique particular?

- (1) Nyaya
- (2) Buddhism
- (3) Vaishesika
- (4) Jaina

18. Which of the following is represented by Noise Index L_{90} ?

- (1) Background noise level
- (2) Average noise level
- (3) Peak noise level
- (4) Threshold noise level

19. Arrange the following chronologically:

- (A) Secondary Education Commission
- (B) Kothari Education Commission
- (C) Establishment of UGC
- (D) Chattopadhyay Commission
- (E) University Education Commission

Choose the **correct** answer from the options given below:

- (1) A, E, B, D, C
- (2) A, D, C, B, E
- (3) E, A, C, B, D
- (4) E, B, D, C, A

20. What among the following can be said about the fallacy of Savyabhicara (irregular middle)?

- (1) It leads to single conclusion
- (2) The hetu in it, is uniformly constant with the major term
- (3) The hetu in it, is only related to the non-existence of the major term
- (4) It leads to different opposite conclusions

21. Which of the following would apply to Qualitative research methodology?

- (A) Empiricism
- (B) Emphasis on description of variables
- (C) Rationalism
- (D) Structured rigid methodology
- (E) Explores experiences, meanings, perceptions and feelings

Choose the **correct** answer from the options given below:

- (1) B, C and D Only
- (2) C, D and E Only
- (3) A and E Only
- (4) A, B and E Only

22. Difficulty in accessing a memory because of the presence of other information is known as _____.

- (1) Decay
- (2) Interference
- (3) Retroactive
- (4) Proactive

23. Which of the following depicts Semantic Memory?

- (1) It is dependent on the person's identity with the past.
- (2) It is a student's general knowledge about the world.
- (3) It is the retention of information about the where and when of life's happenings
- (4) It is the knowledge in the form of skills and cognitive operations.

24. Identify the fallacy in the following:

"Only man has an immortal soul". No woman is a man. Therefore, no woman has an immortal soul".

- (1) Strawman (2) Equivocation
(3) Red Herring (4) Bandwagon argument

25. Which amongst the following is not a Buddhist monastic University?

- (1) Nalanda (2) Takshashila
(3) Vikramshila (4) Odantpuri

26. Match the **List-I** with **List-II**.

List-I (Environmental Treaty/ Convention)		List-II (Focus Area)	
A.	Bonn Convention	I.	Migratory Species Conservation
B.	CITES	II.	Convention on Biological Diversity
C.	Earth Summit	III.	Trades in Endangered Species
D.	Nagoya Protocol	IV.	Sustainable Development

Choose the **correct** answer from the options given below:

- (1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(2) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
(3) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
(4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

27. Match the **List-I** with **List-II**.

List-I (Device)		List-II (Function)	
A.	CPU	I.	stores data temporarily
B.	Monitor	II.	stores data permanently
C.	RAM	III.	Processes Instructions
D.	Hard disk	IV.	Displays visual output

Choose the **correct** answer from the options given below

- (1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(4) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

28. Total gain in Biomass by an animal is known as

- (1) Gross Primary Productivity
(2) Gross Secondary Productivity
(3) Net Primary Productivity
(4) Net Secondary Productivity

29. The form of computing utilized by an email service is

- (1) Distributed (2) Grid
(3) P2P (4) Cloud

30. A shopkeeper purchased 10 kg of Basmati rice at the rate of Rs. 50 per kg and 15 kg of ordinary rice at the rate of Rs. 20 per kg. He mixed the two varieties of the rice and sold it for a gain of 25%. What was his sale price (Rs/kg)?

- (1) Rs 10 per kg (2) Rs 20 per kg
(3) Rs 30 per kg (4) Rs 40 per kg

31. Which of the following are most suitable tools for creating interactive teaching and learning material?

- (A) Genially
(B) Learningapps. org
(C) Interacty
(D) H5P
(E) Educaplay

Choose the **correct** answer from the options given below:

- (1) A, C and E Only (2) B and D Only
(3) A, B, D and E Only (4) A, B, C, D and E

32. What are the three stages of Interpersonal Communication?

- (A) The Phatic Stage
(B) The Plato Stage
(C) The Personal Stage
(D) The proxemics Stage
(E) The Intimate Stage

Choose the **correct** answer from the options given below:

- (1) A, B and C Only (2) A, C and E Only
(3) C, D and E Only (4) B, C and D Only

33. Identify the correct claims in the context of MOOD-AII in IInd figure.

- (A) The middle term appears as predicate in both the premises
(B) It commits fallacy of undistributed middle term
(C) Middle term is distributed in the major premise
(D) Draws negative Conclusion from affirmative premise
(E) Conclusion distributes neither of its terms

Choose the **correct** answer from the options given below:

- (1) C and D Only (2) A, B and E Only
(3) B and C Only (4) B, C and D Only

34. Artificial Wetland is located in

- (1) Keoladeo National Park
(2) Rajaji National Park
(3) Kanha National Park
(4) Manas National Park

35. If 'PEAK' is coded as 'KVZP' and 'FOUR' is coded as 'ULFI', the code of 'THIS' will be:

- (1) GRSH (2) GSRH
(3) SHGR (4) HGRS

36. What is Grapevine communication?
- Corporate Communication
 - Agri base Communication
 - Formal Communication
 - Informal Communication
37. Which among the following is used for judging the significance of several statistical measures, particularly the mean?
- χ^2 -test
 - Z-test
 - Kruskal-Wallis test
 - F-test
38. Arrange the following events related to mass media in the correct chronological sequence.
- Publication of early news papers.
 - Development of Television Broadcasting.
 - Introduction of email.
 - Launch of online news websites.
 - Invention of Radio.
- Choose the **correct** answer from the options given below:
- A, E, B, D, C
 - A, E, B, C, D
 - E, A, C, D, B
 - A, C, B, E, D
39. Choose the next term of the series 6, 18, 108, 1296, _____.
- 31104
 - 15552
 - 30104
 - 23328
40. A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 km/h more than its original speed. If it takes 3 hours to complete total journey, what is the original speed?
- 48 km/h
 - 40 km/h
 - 42 km/h
 - 32 km/h
41. Which of the following is not a security risk?
- Hacking
 - Firewall
 - Phishing
 - Pharming
42. Pick the correct statements pertaining to NEP 2020.
- It promotes multiple entry-multiple exit
 - It promotes English education
 - It promotes multi and inter-disciplinarity
 - It promotes privatization-philanthropic participation in Education
- Choose the **correct** answer from the options given below:
- A, C and D Only
 - B and D Only
 - B, C and D Only
 - A and B Only
43. Which among following definition states that the term to be defined is correctly applied to a given case, if and only if. the performance of specified operations in that case yields a specified result.
- Synonymous definition
 - Operational definition
 - Ostensive definition
 - Quasi-ostensive definition

44. Which of the following is a variant of the Harvard style of referencing?
- APA
 - Chicago
 - IEEE
 - MLA
45. If a researcher decides to have a control group in a 'placebo' design, he/she should have how many groups?
- One group
 - Two groups
 - Three groups
 - Four groups

Comprehension: Read the passage given below and answer the questions that follow:

The most important literary event in the nineteenth century was the emergence of literary prose in all the modern Indian languages. It is not that prose was an invention of this period. Both Sanskrit and Persian had a vast body of prose. Persian prose in the main treated "serious and learned topics"; the purpose of the authors was "to inform and to uplift rather than to amuse", despite the fact that the fantastic adventure of the *Arabian Nights* cycle gave scope to the creative imagination. Sanskrit which developed a simple and dignified prose-style from the time of the composition of the Upanisads, several centuries before Christ, employed prose in various spheres of intellectual and imaginative activities. Following the same tradition. Pali and a few Prakrit languages also developed a rich and vast body of tales and romances.

46. What among the following allowed scope for creative imagination?
- The Ramayana
 - The Arabian Nights
 - The Mahabharata
 - The Panchatantra
47. How is Sanskrit prose described in the passage?
- Simple and dignified
 - Complex and humorous
 - Solely poetic and lyrical
 - Limited to religious themes
48. The most significant literary event in the nineteenth century according to the passage was:
- The invention of poetic forms
 - The development of drama
 - The emergence of literary prose in modern Indian languages
 - The translation of foreign literature
49. What was the purpose of authors writing in Persian prose, as described in the passage?
- To entertain the audience
 - To inform and uplift the readers
 - To promote religious doctrines
 - To provide translations of Sanskrit texts
50. According to the passage, Persian prose deals with
- Serious and learned topics
 - Entertainment to the readers
 - Comedy only
 - Religious practices only

SOLUTIONS

1. (4) Total student enrolled in college A in year 2020 and 2021
 $= 1200 + 1100 = 2300$
 Total student enrolled in college C in year 2022 and 2023
 $= 1200 + 800 = 2000$
 Thus, required % $= \frac{2300 - 2000}{2000} \times 100 = \frac{300}{2000} \times 100 = 15\%$
2. (2) Required ratio $= \frac{3}{8} \times 800 : \frac{5}{12} \times 1200 = 3 : 5$
3. (4) The total number of students enrolled in
 Year 2020 $= 1200 + 900 + 1000 + 700 = 3,800$
 Year 2023 $= 600 + 700 + 800 + 1000 = 3,100$
 Required number $= 3800 - 3100 = 700$
4. (1) Average number of students enrolled in College B over all 4 years $= \frac{900 + 800 + 1000 + 700}{4}$
 $= \frac{3400}{4} = 850$
5. (4) For college B, percentage of students enrolled in BCA in 2021
 $= 100\% - (20\% + 30\%) = 50\%$
 Required difference $= (50\% - 20\%)$ of 800
 $= \frac{30}{100} \times 800 = 240$
6. (4) Video conferencing enhances productivity by enabling real-time collaboration and decision-making. It also promotes eco-friendly practices by reducing travel, lowering carbon emissions, and saving resources.
Lack of Personal Contact is a disadvantage, not an advantage, as virtual meetings cannot fully replicate face-to-face interactions. **Clarity of Audio and Video Always** is incorrect because technical issues like poor internet connectivity can affect quality.
7. (4) $\frac{1}{2} : \frac{1}{3} : \frac{3}{4} = \frac{1}{2} \times 12 : \frac{1}{3} \times 12 : \frac{3}{4} \times 12$
 $= 6 : 4 : 9$
 Share of C $= \frac{9}{19} \times 3800 = \text{Rs. } 1800$
8. (1) Loam soil is ideal for earthworm activity because it has a good balance of sand, silt, and clay, providing optimal moisture and organic matter. Although waterlogged, gleyed soil can still support earthworms in certain conditions due to the presence of organic material. Rendzina soil type is rich in calcium carbonate and organic matter, making it favorable for earthworm activity.
Podzol soils are acidic, nutrient-poor, and low in organic matter, making them unsuitable for earthworm activity. Thus, the correct answer is **A, B, and C Only**.
9. (4) The OSI model follows a layered structure, starting from the Physical Layer (B) and moving up to the Application Layer (E):

Physical Layer (B) – Handles raw data transmission.

Data Link Layer (A) – Manages data frames and error detection.

Network Layer (C) – Handles routing and addressing.

Presentation Layer (D) – Translates and encrypts data.

Application Layer (E) – Provides services for end-users.

Thus, the correct sequence is **(B, A, C, D, E)**.

10. (3) The correct match is:

(A) George Gerbner → (II) Cultivation Theory

Cultivation Theory suggests that prolonged exposure to media, especially television, shapes perceptions of reality.

(B) L.A. Festinger → (I) Cognitive Dissonance Theory

This theory explains how individuals experience discomfort when their beliefs and behaviors are inconsistent, leading them to change their attitudes.

(C) McCombs and Donald L. Shaw → (IV) Agenda Setting Theory

This theory states that the media influences what issues the public considers important by controlling what is prominently covered in the news.

(D) Katz and Blumer → (III) Uses and Gratification Theory

This theory focuses on how individuals actively use media to satisfy their needs, such as entertainment, information, or social interaction.

11. (4) The correct match is :

(A-I, B-IV, C-III, D-II)

(A) Milind College → (I) B.R. Ambedkar

Milind College was established by Dr. B.R. Ambedkar to promote education among marginalized communities.

(B) Rajghat Besant School → (IV) Jiddu Krishnamurti

Rajghat Besant School was founded by Jiddu Krishnamurti, emphasizing holistic education.

(C) Banasthali Vidyapeeth → (III) Hiralal Shastri

Banasthali Vidyapeeth was established by Hiralal Shastri as a premier women's university in India.

(D) Auroville → (II) Sri Aurobindo

Auroville, an experimental township, was inspired by Sri Aurobindo and The Mother, focusing on spiritual and communal living.

12. (3) Among all the given options the correct statements are : B, C, and D, because:

- Multiprogramming allows multiple programs to reside in memory simultaneously, with the CPU switching between them, improving CPU utilization.
- DOS (Disk Operating System) is a single-user, single-tasking operating system, primarily used in early personal computers.

Mainframe systems are designed to support multiple users simultaneously, often handling numerous transactions concurrently in a time-sharing manner.

13. (2) A **scientific hypothesis** must be testable, measurable, and falsifiable through experimentation or observation.

Statement (2) is a valid scientific hypothesis because it can be tested through controlled studies, measuring the impact of almond consumption on heart disease risk.

14. (3) The infamous minutes of 1834 refer to Macaulay's Minute on Indian Education, which played a crucial role in introducing English as the medium of instruction in India. This document promoted Western education and led to the decline of traditional Indian learning systems.

Charles Wood is Associated with the Wood's Despatch of 1854, which emphasized higher education in English. William Bentinck a governor General who implemented Macaulay's recommendations but did not draft the minutes. Charles Grant advocated for English education but was not linked to the 1834 minutes.

15. (2) Cooperative learning is an instructional approach where students work together in small groups to achieve a common goal. Individual and group accountability is a key element, ensuring that each member contributes and learns while being responsible for both their own and the group's progress.

16. (4) The Episodic Buffer, proposed by Baddeley (2000), is a component of working memory that integrates information from multiple sources. It acts as a temporary storage system that links information from:

- Phonological Loop (verbal and auditory information)
- Visuospatial Sketchpad (visual and spatial data)
- Long-Term Memory (previously stored knowledge)

17. (2) The **Buddhist epistemological tradition**, particularly from the **Sautrantika and Yogacara schools**, upholds **perception (pratyakṣa) as an unerring knowledge of the unique particular (svalakṣaṇa)**. According to Buddhism, perception is direct, non-conceptual, and corresponds precisely to reality without any distortion.

Buddhists classify objects into:

Svalakṣaṇa (Unique Particulars) – Directly perceived, momentary, and real.

Sāmānyalakṣaṇa (Universals or Generalities) – Conceptually constructed and not directly real.

18. (1) The Noise Index L_{90} represents the noise level exceeded for 90% of the measurement period, commonly used to determine background noise levels in an environment. It helps assess the lowest consistent noise level in an area. Whereas

19. (3) The correct chronological order is:

University Education Commission (E) - 1948-49

- Led by Dr. S. Radhakrishnan, focused on higher education reforms.

Secondary Education Commission (A) - 1952-53

- Also known as the Mudaliar Commission, focused on secondary education structure.

Establishment of UGC (C) - 1956

- University Grants Commission was set up to regulate and fund higher education institutions.

Kothari Education Commission (B) - 1964-66

- Introduced the 10 + 2 + 3 education system and emphasized national education policies.

Chattopadhyay Commission (D) - 1983

- UGC Review Committee under Prof. D.P. Chattopadhyay, reviewed UGC's functioning.

20. (4) The fallacy of Savyabhicāra (Irregular Middle) occurs when the middle term (hetu) is inconsistent in its relation to the major term (sādhya). This inconsistency leads to multiple or contradictory conclusions, making the inference unreliable.

For example, if the middle term (hetu) is present in both cases where the major term is present and absent, it fails to establish a consistent relationship, leading to opposite or conflicting conclusions.

21. (4) Qualitative research methodology focuses on understanding human experiences, meanings, and perceptions rather than numerical data. It emphasizes empiricism (A) by relying on real-world observations, describes variables in detail (B) rather than measuring them quantitatively, and explores subjective experiences (E) to interpret human behavior.

Rationalism (C) is incorrect because qualitative research does not rely on logical reasoning alone but on interpretive analysis. Structured and rigid methodology (D) is associated with quantitative research, whereas qualitative methods are flexible and adaptive.

22. (2) Interference occurs when the presence of other information disrupts memory retrieval. It happens in two forms: proactive interference, where old information hinders new memory recall, and retroactive interference, where new learning affects the recall of old information.

Decay (Option 1) is incorrect as it refers to memory loss over time due to disuse rather than interference. Retroactive (Option 3) and Proactive (Option 4) are types of interference, but the general term covering both is simply "interference."

23. (2) Semantic memory refers to general world knowledge, including facts, concepts, and meanings that are not tied to personal experiences (e.g., knowing that Paris is the capital of France).

While the other Options Are Incorrect because Statement (1) Describes episodic memory (personal experiences). Statement (2) also refers to episodic memory and statement (3) Describes procedural memory (e.g., riding a bike).

24. (2) The fallacy in the given argument is Equivocation, which occurs when a word is used in two different senses within the same argument, leading to a false conclusion.

In this case, the term "man" is used ambiguously:

- In the first statement, "Only man has an immortal soul", "man" could mean humankind (all humans).
- In the second statement, "No woman is a man", "man" is interpreted as male (as opposed to female).

25. (2) Takshashila (Taxila) was an ancient center of learning but not a Buddhist monastic university. It was a prominent educational hub in ancient India, attracting scholars from various disciplines, including Vedic studies, law, medicine, and military science.

Nalanda (Option 1), Vikramshila (Option 3), and Odantpuri (Option 4) were all well-known Buddhist monastic universities, established to promote Buddhist learning and philosophy.

26. (2) The correct match is:

Bonn Convention → (I) Migratory Species Conservation

Focuses on the protection of migratory wildlife across borders.

CITES → (III) Trades in Endangered Species

Regulates international trade to prevent species extinction.

Earth Summit → (IV) Sustainable Development

1992 Rio Summit addressed environmental and developmental challenges.

Nagoya Protocol → (II) Convention on Biological Diversity

Aims at fair and equitable sharing of genetic resources.

27. (2) The correct match is : **A-III, B-IV, C-I, D-II**

The **Central Processing Unit (CPU)** is the brain of the computer, responsible for executing instructions and processing data.

The **Monitor** is an output device that visually displays information processed by the computer.

Random Access Memory (RAM) is a volatile memory that temporarily stores data and instructions for quick access by the CPU.

The **Hard Disk (HDD/SSD)** is a non-volatile storage device that permanently stores data, applications, and the operating system.

28. (2) Gross Secondary Productivity (GSP) refers to the total gain in biomass by an animal through consumption of food, before accounting for losses due to respiration and excretion. It represents the total energy assimilated by heterotrophs in an ecosystem.

Gross Primary Productivity (Option 1) applies to plants and measures the total energy captured via photosynthesis. Net Primary Productivity (Option 3) is the energy left in plants after respiration. Net Secondary Productivity (Option 4) refers to the energy available for growth and reproduction after subtracting energy lost through respiration.

29. (4) Email services operate on cloud computing platforms, enabling users to send, receive, and store emails remotely. Cloud-based email services provide scalability, accessibility, and data redundancy, ensuring efficient communication.

30. (4) Cost of 10 kg basmati rice = $10 \times 50 = \text{Rs.} 500$

Cost of 15 kg ordinary rice = $15 \times 20 = \text{Rs.} 300$

Total cost = Rs. 800

Total rice = $10 + 15 = 25 \text{ kg}$

Cost per kg = $\frac{800}{25} = \text{Rs.} 32 / \text{kg}$

Thus, S.P = $32 \times \left(1 + \frac{25}{100}\right) = 32 \times 1.25 = \text{Rs.} 40 \text{ per kg}$

31. (4) All listed tools are widely used for creating interactive teaching and learning materials, offering features like quizzes, games, presentations, and multimedia content.

32. (2) The three stages of interpersonal communication are **Phatic Stage, Personal Stage**, and Intimate Stage. The Phatic Stage involves small talk and casual greetings to establish social interaction. The Personal Stage allows deeper conversations, fostering connections. The Intimate Stage includes close, emotionally connected communication.

The **Plato Stage (Option B)** is not a recognized stage in interpersonal communication. The **Proxemics Stage (Option D)** refers to the study of personal space rather than a communication stage. Thus, the correct answer is **A, C, and E Only**.

33. (2) Mood A I I means:

I. Major premise is an A-proposition ("All ... are ...").

II. Minor premise is an I-proposition ("Some ... are ...").

III. Conclusion is an I-proposition ("Some ... are ...").

Figure 2 means the middle term M is the predicate in both premises.

A typical layout, then, is:

All P are M (A)

Some S are M (I)

∴ Some S are P (I)

(A) "The middle term appears as predicate in both premises." - True. By definition of Figure 2, M is indeed the predicate term in both premises.

(B) "It commits fallacy of undistributed middle term." - True. In the premise "All P are M," P (the subject) is distributed, but M (the predicate) is not. In "Some S are M," neither S nor M is distributed. Hence M is undistributed in both premises, violating the requirement that the middle term be distributed at least once.

(C) "Middle term is distributed in the major premise." - False. As noted, in an A-proposition (All P are M), the subject P is distributed, not the predicate M. So M is not distributed here.

(D) "Draws negative conclusion from affirmative premise." - False. The conclusion in A I I is "Some S are P" (an I-proposition, which is affirmative), so there is no negative conclusion at all.

(E) "Conclusion distributes neither of its terms." - True. An I-proposition ("Some S are P") distributes neither subject nor predicate.

Hence the correct set of claims is A, B, and E.

34. (1) Keoladeo National Park in Rajasthan is known for its artificially managed wetland that serves as a crucial habitat for migratory birds. The wetland was created by human intervention, primarily for waterfowl conservation and birdwatching.

Rajaji National Park (Option 2) is primarily a forested area known for elephants. Kanha National Park (Option 3) is famous for its tiger reserve and dense forests. Manas National Park (Option 4) is a UNESCO-listed biosphere reserve with natural wetlands, but they are not artificial.

35. (2) As,

Opposite letters →

P	E	A	K
↓	↓	↓	↓
K	V	Z	P

And,

Opposite letters →

F	O	U	R
↓	↓	↓	
U	L	F	I

Similarly,

Opposite letters →

T	H	I	S
↓	↓	↓	↓
G	S	R	H

36. (4) Grapevine communication refers to an **informal mode of communication** within an organization, often spreading information, rumors, or gossip through personal conversations

and social interactions. It is spontaneous, unstructured, and not officially recognized but plays a crucial role in information flow.

Corporate communication (Option 1) refers to structured communication within a business. Agri-based communication (Option 2) relates to communication in the agricultural sector. Formal communication (Option 3) follows an official, structured format within an organization.

37. (2) The Z-test is used to determine whether there is a significant difference between the means of two large sample sizes when the population variance is known. It is particularly useful for testing hypotheses about population means and proportions. Since the question asks about judging the significance of several statistical measures, particularly the mean, the Z-test is the most appropriate choice.

38. (2) The chronological sequence of mass media developments is as follows:

A - Publication of early newspapers (17th century) : Newspapers were one of the earliest forms of mass communication, widely circulated since the 1600s.

E - Invention of Radio (Late 19th to early 20th century) : Radio was invented in the late 1800s and became a major medium of mass communication in the early 20th century.

B - Development of Television Broadcasting (Mid-20th century) : Television broadcasting started in the 1920s and became widespread by the 1950s.

C - Introduction of Email (1970s) : Email was first developed in the early 1970s as part of ARPANET, the precursor to the internet.

D - Launch of Online News Websites (1990s) : Online news platforms emerged in the late 1990s with the rise of the internet.

39. (1) 6, 18, 108, 1296, 31104
-

40. (3) Let the original speed of train = x km/hr

$$\text{Time taken for 63 km} = \frac{63}{x} \text{ hr}$$

$$\text{Time taken for 72 km} = \frac{72}{x+6} \text{ hr}$$

According to question,

$$\Rightarrow \frac{63}{x} + \frac{72}{x+6} = 3$$

$$\Rightarrow \frac{63x + 378 + 72x}{x(x+6)} = 3$$

$$\Rightarrow 135x + 378 = 3x^2 + 18x$$

$$\Rightarrow 3x^2 - 117x - 378 = 0$$

$$\Rightarrow 3x^2 - 126x + 9x - 378 = 0$$

$$\Rightarrow 3x(x-42) + 9(x-42) = 0$$

$$\Rightarrow (3x+9)(x-42) = 0$$

So, $x = 42$ (possible value).

41. (2) A firewall is a security mechanism that protects networks by controlling incoming and outgoing traffic based on security rules. It is not a security risk but rather a protective measure against cyber threats.

Hacking is an unauthorized attempt to exploit a system, making it a security risk. Phishing involves fraudulent attempts to obtain sensitive information, posing a major security threat. Pharming redirects users to fake websites to steal credentials, which is also a security risk.

42. (1) The National Education Policy (NEP) 2020 supports a multiple entry-multiple exit system, allowing students flexibility in their education pathways. It also promotes multi and interdisciplinary, ensuring holistic learning. Additionally, NEP 2020 encourages privatization and philanthropic participation in education to enhance accessibility and quality.

NEP 2020 does not emphasize English as the sole medium of instruction but rather supports multilingualism and regional languages, making option B incorrect.

43. (2) An operational definition specifies the exact procedures or operations needed to measure or identify a concept. It ensures that a term is correctly applied only if a particular process leads to a specified result, making it essential in scientific and empirical research.

A synonymous definition explains a term using words with the same meaning. An ostensive definition defines by pointing to examples, and a quasi-ostensive definition expands ostensive definitions with additional context, making them incorrect choices.

44. (1) The APA (American Psychological Association) referencing style is considered a variant of the Harvard style as it follows an author-date citation system similar to Harvard. It is widely used in social sciences and education.

The **Chicago style** allows both author-date and footnote referencing, making it different. The IEEE style is numerical and used in technical fields, while **MLA (Modern Language Association)** is primarily for humanities, relying on parenthetical citations, distinguishing them from Harvard.

45. (2) A placebo-controlled design requires at least two groups:

Experimental group – Receives the actual treatment.

Control group – Receives a placebo (inactive substance) to compare effects. The official answer is three group but the correct answer is two groups.

46. (2) The passage states that The Arabian Nights allowed scope for creative imagination. It highlights the fantastic adventures in The Arabian Nights as an example of imaginative prose.

47. (1) The passage states that Sanskrit developed a simple and dignified prose style from the time of the composition of the Upanishads.

48. (3) The passage states that the most important literary event in the nineteenth century was the emergence of literary prose in all the modern Indian languages.

49. (2) The passage states that Persian prose mainly dealt with “serious and learned topics” and that the purpose of the authors was “to inform and to uplift rather than to amuse.”

50. (1) The passage explicitly states that Persian prose in the main treated “serious and learned topics.”

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- June 2019 Session - 3 Papers
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